Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **Java/Python-adjacent tools**, SQL-backed services, REST APIs, automated tests, troubleshooting documentation, and reliable production software.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*

- Built **SQL-backed platform workflows** for 500+ users, translating requirements into APIs, validation logic, monitored jobs, technical documentation, release notes, and maintainable delivery through reviewed changes and support notes.
- Improved **backend reliability and performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with diagnostic handoffs, reviews, and runbooks.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*

- Engineered **Python API and ETL services** using SQL, AWS Glue, and S3, improving response time by **30%** across integrations, release reviews, operational handoffs, troubleshooting, documentation, and supportability.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** while reducing release regressions during deployments, reviews, and checks.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Developed **Python services and React tooling** handling 10,000+ daily requests, implementing file ingestion, REST endpoints, automated checks, documented fixes, and sprint-ready production workflows with reviews and testing.
- Debugged **production workflow issues** by reproducing OCR, persistence, and execution defects, documenting regression cases, reviewing fixes, and cutting instructor feedback turnaround by **60%** across validation cycles and releases.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*

- Built **reusable application modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, workflows, integrations, releases, debugging, validation, monitoring, support, maintenance, and reliability.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, validation notes, and shared-environment handoffs across staging workflows.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*

- Built **systems evaluation pipelines** across 100,000+ trajectory records, producing metrics, logs, reports, and reproducible artifacts for model behavior debugging, experiment comparison, validation, diagnostics, reviews, and reports.
- Documented **edge cases and integration behavior** for researchers, converting exploratory experiments into repeatable test scenarios with traceable outputs, clearer handoffs, diagnostics, reviewable conclusions, validation notes, and reports.

Skills

Languages: Java, Python, C++, C#, JavaScript, TypeScript, SQL, Bash
Backend: REST APIs, JSON, XML familiarity, Flask, FastAPI, Spring Boot awareness, service design, reusable components
Data: PostgreSQL, MySQL, SQL Server, MongoDB, RDBMS concepts, query tuning, validation, reporting
Systems: data structures, algorithms, distributed-systems concepts, Linux, debugging, troubleshooting, performance analysis
Delivery: Git, Docker, Jenkins, CI/CD, PyTest, Mockito, automated testing, code reviews, documentation, flowcharts

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services, TypeScript interfaces, Docker workflows, Git collaboration, and testing.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ scheduling** with metric calculation and visualization, strengthening algorithms, C++ fundamentals, runtime reasoning, debugging discipline, scheduler comparisons, and performance analysis.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized Python REST API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting, integration notes, and automated tests.